25 September 1974
SUPERSEDING
MIL-M-21734B(WEP)
1 JULY 1963

MILITARY SPECIFICATION

MICROPHONE, DYNAMIC, OXYGEN MASK M-94B/A

This specification has been approved by the
Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - The microphone covered by this specification shall be noise cancelling and suitable for use with standard Navy oxygen masks.

*1.2 Classification - The Microphone, Dynamic, Oxygen Mask M-94B/A shall be of one type and shall consist of the following items:

Item	Appl. Para.
Transmitter Assembly	3.4.1.3.1
Bracket Assembly	3.4.1.3.2
Cord Assembly	3.4.1.3.3
Connector, Type U-173/U	3.4.1.3.4

When a complete Dynamic Microphone is not required, any one item or combination of items may be procured accordingly as specified in the procurement document (See 6.3).

2. APPLICABLE DOCUMENTS

*2.1 General - The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein (See 6.4(4)).

SPECIFICATIONS

Military

MIL-P-642	Plugs, Telephone
MIL-C-5898	Cord, Electrical, WD-34/U
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of

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MIL-N-18307 Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment

MIL-Q-9858 Quality Control System Requirements

Naval Air Systems Command

AR-34 Failure Classification for Reliability Testing, General Requirements for

STANDARDS

Military

MIL-STD-781 Reliability Tests, Exponential Distribution

MIL-STD-794 Parts and Equipment, Procedures for Packaging and Packing of

*2.2 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Publications and Forms Center, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 Qualification - The microphone furnished under this specification shall be a product which has been tested, has passed the qualification tests specified herein, and has been listed on, or approved for listing on, the applicable qualified products list.

*3.2 Parts and Materials - In the selection of parts and materials, fulfillment of design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-E-5400.

(2) Nonrepairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable.

3.2.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

3.3 Design and Construction -

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3.3.1 General - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, unless otherwise specified herein.

3.3.2 Service Conditions - The microphone shall operate satisfactorily under any of the environmental and mechanical service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 1 equipment. The requirement of Specification MIL-E-5400 for sand and dust and for explosive tests shall not be required.

3.3.2.1 Altitude - The microphone shall reduce the effects of changes in altitude. At 40,000 feet, the frequency response of any given microphone shall not change from its ground level response by more than 3 db.

3.3.3 Standard Conditions - The following conditions shall be used as a basis to establish normal performance and for making laboratory bench tests:

Temperature	Room ambient 30° (±10°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity

3.3.4 Weight - The weight of the microphone shall not exceed 1.5 ounces.

3.3.5 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-E-5400.

3.3.6 Operational Stability - The microphone shall operate with optimum performance for 500 hours, continuously or intermittently.

3.3.7 Reliability in Mean Time Between Failures - The microphone shall have a minimum of 500 hours of mean (operation) time between failures when tested and accepted under the requirements of 4.7.

3.3.8 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with the requirements of Specification MIL-N-18307.

3.3.9 Provisions for Maintenance and Field Testing - The transmitter assembly is a sealed unit. When determined to be defective, disposal procedures will be followed. Provisions for field testing of the microphone shall be as specified in Specification MIL-E-5400.

3.3.10 Insulating and Case Materials - Insulating and case materials shall comply with the following requirements, as applicable.

3.3.10.1 Molded Plastic-Material - Molded plastic-material shall be of such character and quality that it will withstand all service conditions, including test conditions outlined in Section 4 of this specification, without evidence of affecting the performance requirements of the microphone.

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3.3.10.2 Insulating and Impregnating Compounds - All such compounds including varnishes, waxes and the like, shall be suitable in every respect for each particular application. Under all service conditions, including test conditions outlined in Section 4 of this specification, the compound shall adequately preserve the dielectric strength of the insulation to which it is applied. A compound shall not, either in the state of its original application or as a result of aging under severe service conditions, show any injurious effect upon the insulation it is designed to protect, and shall not cause corrosion or deterioration of adjacent metal or plastic parts.

3.3.10.3 Cementing Compound - Any cementing compound used in the construction of the microphone shall be of such a character and quality that it will withstand all service conditions including test conditions outlined in Section 4 of this specification without evidence of loosening or otherwise affecting the performance requirements of the microphone.

3.3.11 Cleaning and Finishing of Metal Parts -

3.3.11.1 Cleaning - Where cleaning operations on metal parts are not specified in detail, they shall be in accordance with the best suitable commercial practice.

3.3.11.2 Aluminum Alloys - Where aluminum alloy parts are employed they shall be cleaned by immersion in a hot caustic or other suitable solution until etched, rinsed thoroughly in clean cold water, immersed in nitric acid until the remaining caustic is neutralized, and thoroughly rinsed in cold, then hot water. Anodic treatment is not prohibited, but masking shall be employed where necessary to retain the quality of the original metal at areas where good electrical contact is necessary.

3.3.11.3 Other Non-Ferrous Metals and Alloys - Copper, Brass, Bronze, Berryllium-Copper and other similar materials shall be plated or otherwise suitably finished or enclosed to provide protection against corrosion and deterioration. The kind of plating or finish applied shall depend upon the characteristics of the individual part and its use in the microphone. The plating or finish used for each individual part shall be such as to provide good appearance, corrosion and deterioration resistance, and long life for that part. Various problems such as soldering, fitting, assembly, reaction with adjacent parts and the like, must be considered in the selection of the plating or finish.

3.3.11.4 Soldering Flux - Only resin, or resin and alcohol, may be used as a flux in the assembly of this equipment or any part thereof, except that an acid flux may be used on all-metal, non-electrical sub-assemblies provided that they are thoroughly and completely cleaned of all traces of the acid immediately after the soldering operation has been completed.

3.4 Detail Requirements - The design and assembly of the microphone shall be such that the microphone is inherently stable as to mechanical and electrical characteristics. The detailed mechanical and electrical design of the microphones shall be accomplished by the contractor, subject to the requirements of this specification. The requirements are

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detailed herein only to the extent considered necessary to obtain the desired mechanical and electrical characteristics and performance, and permanence of the same.

3.4.1 Microphone, Dynamic, Oxygen Mask M-94B/A - The microphone shall meet the following requirements.

3.4.1.1 Function - The microphone shall convert normal speech input toelectrical signals suitable for connection, via a preamplifier, into the speech amplifiers and/or modulators of radio or audio equipment.

3.4.1.2 Form Factor - The microphone shall have the dimensions shown in Figures 1 and 2.

3.4.1.3 Contents of the Microphone - The microphone shall be an assembly composed of the following:

*3.4.1.3.1 Transmitter Assembly - The transmitter assembly, as shown in Figure 1, shall be individually interchangeable with components previously manufactured and shall be compatible with the required amplifier. The transmitter assembly shall not weigh in excess of 0.60 ounce and shall provide the performance specified in 3.5 to3.5.7 inclusive. The transmitter assembly shall be secured to the bracket assembly with two screws and the terminals shall be in accordance with Figure 2.

3.4.1.3.2 Bracket Assembly - The bracket assembly shall consist of an insulating terminal block and adjustable microphone support having the general configuration shown in Figure 1. The bracket assembly shall not weigh in excess of 0.30 ounce. Terminals to which the cord assembly is attached shall be protected to prevent the possibility of ground or shorting by suitable means such as an insulating washer, when installed in smoke masks.

*3.4.1.3.3 Cord Assembly - The cord shall be Type WD-34/U in accordance with Specification MIL-C-5898 and Figure 9.

*3.4.1.3.3.1 Anchorage - The cord shall be anchored at the microphone bracket. The anchorage shall withstand a tension test of 10 pounds with the wires not connected.

*3.4.1.3.4 Connector - A U-173/U plug, equivalent to PJ-292 of Specification MIL-P-642, shall be provided for the cord assembly. The plug shall withstand a tension test of 20 pounds.

3.4.2 Associated Equipment - The microphone shall operate with the following associated equipment which shall not be supplied as part of the equipment.

3.4.2.1 Oxygen Mask - The microphone shall be capable of installation in the Navy type MS22001 Oxygen Mask, Smoke Mask, or other Military equipment which will seat the type ANB-M-C1 microphone or a similar microphone.

3.4.2.2 Interconnecting Box J-1013/AIC - The microphone shall be capable of operation with the transistorized microphone amplifier con-

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tained in the Interconnection Box J-1013/AIC, or equivalent such as Amplifier, Audio Frequency AM-3597/A.

3.5 Performance -

3.5.1 Sensitivity - The voltage output of the microphone at 1000 cps, with a sound pressure level input of 115 db above 0.0002 dyne per sq. cm., shall fall within the limit 51.5 ±3 db above 1 microvolt. The voltage output shall be measured across a 5 ohm load, as shown in Figure 3. The sensitivity of the microphone shall not vary by more than 3 db after being subjected to any, or all, of the service conditions specified herein except salt spray. At an altitude of 10,000 feet the sensitivity shall not fall below the sensitivity at sea level by more than 3 db.

3.5.2 Frequency Response - The frequency response of the microphone, when measured as specified in 4.5.3.3, compared to the response of 51.5 ±3 db at 1000 Hz shall be within the limits shown in Figure 6. However, the difference between any two successive readings 100 Hz apart up to 1000 Hz and 250 Hz apart above 1000 Hz shall not exceed 2.5 db under all standard and service conditions. The response shall fall off at a rate of not less than 3 db per octave below 300 Hz and 6 db per octave above 4000 Hz.

3.5.3 Distortion - At any frequency between 350 and 4000 Hz, when the sound pressure at the face of the microphone is 125 db above 0.0002 dyne per square centimeter, the harmonic distortion in the microphone shall not exceed 4% under standard conditions or 5% after being subjected to service conditions.

3.5.4 Linearity - The output voltage of the microphone shall be a linear function with respect to the input sound pressure in the range of 80 to 125 db above 0.0002 dyne per square centimeter and distortion resulting from non-linearity shall not exceed the limits of 3.5.3.

3.5.5 Impedance - The impedance of the microphone shall be within 5 ±1 ohms between 300 and 4000 Hz under standard or service conditions and shall be measured in accordance with 4.5.3.4.

3.5.6 Pressure Equalization - The response of the microphone at 1000 Hz shall not vary by more than 4 db from the reference established at 15,000 feet while the microphone is subjected to a pressure increase corresponding to a descent from an altitude of 15,000 feet (16.7 inches of mercury) to sea level (30.0 inches of mercury) at a rate of 5000 feet per minute.

3.5.7 Stray Magnetic Field - The stray magnetic field of the microphone shall not cause more than a 3 degree deflection of a magnetic compass when tested as required in 4.5.12.

3.5.8 Noise Immunity Characteristic - Under standard conditions the signal-to-noise ratio shall be at least 15 db in favor of the signal for noises of the types encountered in aircraft, boats, tanks, and other vehicles when measured in accordance with 4.5.3.6.

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3.5.9 Salt Spray - Immediately after the salt spray test (4.5.6) check the microphone for satisfactory operation.

3.5.10 Temperature Shock - The response of the microphone shall not vary more than 3 db from its initial response at any frequency between 300 and 4000 Hz after being subjected to the temperature cycling test specified in 4.5.7 and the resultant signal-to-noise ratio shall not be reduced by more than 2 db.

3.5.11 Impact - The microphone shall withstand the impact test specified in 4.5.13 with no resultant structural failure or other defects. The frequency response shall not show a change greater than 3 db from its original response at any frequency between 300 and 4000 Hz and the resultant signal-to-noise ratio shall not decrease by more than 2 db.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

*4.1.1 Classification of Tests - The sampling, inspection and test of Microphone, Dynamic, Oxygen Mask M-94B/A, shall be classified as follows:

(1) Qualification Tests - Qualification Tests are those tests accomplished on samples submitted for qualification as a satisfactory product.

(2) Acceptance Tests - Acceptance Tests are those tests accomplished on microphones manufactured and submitted for acceptance under the contract.

(a) Individual Tests - Individual Tests are those tests conducted on each microphone submitted for acceptance under the contract.

(b) Sampling Tests - Sampling Tests are those tests accomplished on a quantity of microphones as approved by the Naval Air Systems Command which have previously passed Individual Tests.

(3) Life Tests - Life Tests shall be conducted by the contractor at his plant. These tests shall be started as soon as practicable on an early production microphone.

*4.2 Qualification Tests - Qualification Tests are to be made on samples submitted for qualification as satisfactory production units. These tests shall be conducted at a laboratory and under conditions satisfactory to the Naval Ammunition Depot, Crane, Indiana and shall consist of all the tests listed in 4.4, Table II. Twenty units shall be submitted

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for qualification tests. Modification required as a result of these tests shall be incorporated in the microphones prior to resubmission for qualification. Requests for authorization of qualification tests shall be addressed to the Naval Ammunition Depot, Crane, Indiana 47522, Code 3072. (See 6.2).

*4.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the required tests under the supervision of the cognizant Government Inspector. Contractors not having laboratory testing facilities satisfactory to the Government for sampling tests shall engage the services of a commercial testing laboratory acceptable to the NAD, Crane, Indiana. The contractor shall furnish test reports, in duplicate, via the cognizant Government Inspector, showing quantitative results for all tests required by this specification, and signed by an authorized representative of the contractor or laboratory as applicable. Acceptance or approval of material during course of manufacture shall in no case be construed as a guaranty of the acceptance of the finished product. Acceptance of the finished product shall be upon a lot basis after satisfactory completion of all required tests. Each lot shall be composed of microphones which have successfully passed the individual tests. Acceptance tests shall consist of the following tests.

4.3.1 Individual Tests - Each microphone submitted for acceptance shall be subjected to the individual test to determine compliance with requirements of material, workmanship, and operational adequacy. As a minimum, each microphone shall be subjected to the tests specified in Table II.

4.3.2 Sampling Tests - Sampling Tests shall be conducted on samples from each lot as specified in the following paragraph.

4.3.2.1 Sampling Instructions - One microphone from the first 10 production microphones, and thereafter two from each lot of 100, or fraction thereof, as selected by the cognizant Government Inspector shall be given the specified sampling tests of Table II.

*4.4 Tests Required - The microphone sample shall be subjected to the tests listed in Table II and any other tests deemed necessary by the Naval Air Systems Command or the cognizant Government Inspector to determine compliance of the microphones with the requirements of this specification.

TABLE II

Tests	Test Procedure	Qualification Tests	Individual Tests	Sampling Tests
Examination of Product	4.5 3.1	X	X	X
Operational Test	4.5.3.2	X	X	X
Frequency Response	4.5.3.3	X		X
Impedance Measurement	4.5.3.4	X	X	X
Response Measurements	4.5.3.5		X	
Signal-To-Noise Ratio	4.5.3.6	X	X	X
Stray Magnetic Field Measurement	4.5.12	X		X
Temperature Extremes	4.5.5	X		
Temperature Cycling	4.5.7	X		X
Pressure Cycling	4.5.4.1	X		

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TABLE II Cont.

Tests	Test Procedure	Qualification Tests	Individual Tests	Sampling Tests
Pressure Equalization	4.5.4.2	X		X
Vibration	4.5.11	X		*
Impact	4.5.13	X		*
Humidity	4.5.8	X		
Distortion	4.5.9	X		X
Linearity	4.5.10	X		
Salt Spray	4.5.6	X		
Cord Test	4.5.14	X		
Life Test	4.6	X	**	
Reliability Assurance***	4.7	X		

* These tests shall be performed on the sample microphone from the first 10 production units and on the sample units for every fifth lot. Microphones subjected to this test shall not be submitted for acceptance under the contract.

** Life tests shall be made as per schedule of 4.6.3 and not on Individual or Sampling Test basis.

*** Reliability Assurance Tests shall be made per schedule of 4.7.1 and 4.7.2.

4.5 General Test Procedure -

4.5.1 Test Circuit Apparatus - Measuring and testing devices shall be calibrated and maintained in accordance with Specification MIL-Q-9858. Test apparatus shall be assembled and connected as shown in Figures 3 and 4 and shall consist of equipment described as follows:

4.5.1.1 Audio Oscillator - The audio oscillator shall have a frequency range of at least 100 to 10,000 Hz, shall have a high degree of stability in both output voltage and frequency and shall have a waveform distortion of less than 2 percent.

4.5.1.2 Vacuum Tube Voltmeter - The vacuum tube voltmeters (VTVM) used in the test circuit shall have a flat frequency response (plus or minus 1 db) from 100 to 10,000 Hz and shall be capable of measuring voltages from 0.0001 to 10 volts rms.

4.5.1.3 Sound Source - The sound source shall be a Western Electric 555 speaker, or equivalent. The distortion in the output of the sound source shall be such that the second harmonic is at least 35 db, the third at least 40 db, and the fourth at least 45 db below the fundamental.

4.5.1.4 Baffle - A wooden ring shall be mounted on the front of the sound source, as shown in Figure 4, in a plane parallel with the face of the microphone.

4.5.1.5 Coupler - A brass coupler, of the dimensions shown on Figure 5 shall be mounted concentrically on the sound source as shown on Figure 4.

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4.5.1.6 Base - The test equipment shall be mounted on a rigid base of suitable design.

4.5.1.7 Distortion Analyzer - The distortion analyzer shall be of the wave analyzer or total distortion type. The frequency range shall be at least 100 to 10,000 Hz with high selectivity and an input voltage range from .001 to 10 volts.

4.5.1.8 Signal-To-Noise Ratio Test Apparatus - The Signal-to-Noise Ratio Test Apparatus shall be similar or equal to that detailed in Figure 7 and the test circuit similar or equal to that shown in Figure 8.

(1) Noise Source - By means of a suitable test apparatus and circuit, Figures 7 and 8, provision shall be made for the production of the noise spectrum, Table III, at an rms sound pressure of 115 db above a reference level of 0.0002 dyne per square centimeter as measured at the microphone diaphragm. The "noise" source shall be a loudspeaker capable of producing a sound pressure of 115 db above a reference level of 0.0002 dyne per square centimeter at the rear as well as the front of the microphone and the loudspeaker shall be located as far away from the microphone as practicable. The calibrating microphone shall be mounted in the test rig, in the box, 1/4 inch from the mouth of the signal speaker which shall be short-circuited during adjustments of the noise spectrum.

(2) Signal Source - By means of the apparatus as set up previously, provision shall also be made for the production of the following signal spectrum, Table IV, at a rms sound pressure of 115 db above a referenced level of 0.0002 dyne per square centimeter. The "signal" source shall be the driver unit. The calibrating microphone shall be positioned 1/4 inch directly in front of the driver unit.

TABLE III

NOISE SPECTRUM

Frequency cps	Sound pressure in db relative to sound pressure at 130 Hz Ref.
40	0
70	0
130	0
300	-5
600	-9
1000	-13

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TABLE III Cont.

Frequency Hz	Sound pressure in db relative to sound pressure at 130 Hz Ref.
2000	-17
3000	-19
4000	-21

TABLE IV

SIGNAL SPECTRUM

Frequency Hz	Sound pressure in db relative to sound pressure at 130 Hz Ref.
130	0
300	+7
600	+8
1000	+5
2000	-3
3000	-7
4000	-9

4.5.2 Test Calibration Procedure - The condenser microphone shall be mounted as shown in Figure 4 with an air gap of 1/4 inch separating the coupler and the microphone grid. The microphone axis shall coincide with the center line of the mouth of the sound source. After having determined the electrical input to sound source required to yield a constant sound pressure level of 115 db re 0.0002 dyne per square centimeter, the condenser microphone shall be removed and the test microphone shall be mounted as shown in Figure 4. The geometric center line of the grid shall coincide with the center line of the mouth of the sound source.

4.5.3 Test Methods - Tests required by this specification shall be conducted in accordance with the following precedures:

4.5.3.1 Examination of Product - Each microphone shall be examined carefully to determine conformance with this specification with respect to material, weight, dimensions, construction and workmanship.

4.5.3.2 Operational Test - Each microphone shall be operated. The period of operation shall be long enough to permit the temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory operation (include Impedance Measurement 4.5.3.4, and Response Measurement 4.5.3.5).

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4.5.3.3 Frequency Response - Response measurements shall be made between 300 and 4000 Hz at intervals of 100 Hz up to 1000 Hz and at intervals of 250 Hz above 1000 Hz with the microphone mounted 1/4 inch from the calibrated close sound source as shown in Figure 3. The response shall be recorded in db relative to 1 microvolt.

4.5.3.4 Impedance Measurement - Apply an audio frequency between 300 and 4000 Hz from an oscillator to the microphone connector in series with a 10 ohms ±1% resistive load. The gain of the oscillator shall be adjusted so that the voltage drop across the resistive load shall be 10 millivolts ±2%. The voltage drop across the microphone, in millivolts, shall be taken as the impedance of the microphone measured in ohms.

4.5.3.5 Response Measurements (Individual Tests) - Each microphone shall be subjected to response tests at no less than 6 approved frequencies between 300 and 4000 Hz. The specified frequencies to be used will be determined by the procuring activity on the basis of tests made and information gained in the initial stages of production. The method of testing and the selection of frequencies shall be such that they will truly indicate the over-all response characteristics and output of the microphone.

4.5.3.6 Measurement of the Signal-to-Noise Ratio - The microphone under test shall be mounted in the test rig by the same means described in 4.5.2 with the front grid of the microphone 1/4 inch from the mouth of the signal speaker. (A baffle will not be used). The noise spectrum (Table III) shall be applied and the microphone output measured. The noise shall be removed and the signal spectrum (Table IV) applied. The output of the microphone shall be measured. The signal-to-noise ratio shall be the quotient of these measured values. Refer 3.5.8.

4.5.4 Altitude -

4.5.4.1 Pressure Cycling - The microphone shall be subjected to 5 varying pressure cycles. Each pressure cycle shall consist of 30 minutes at 3.4 inches of mercury (approximately 50,000 feet) and 30 minutes at 30 inches of mercury (approximately sea level). The pressure transition shall be approximately 5000 feet per minute. The response characteristic of the microphone shall then be measured and shall not show a decrease in output in excess of 4 db.

4.5.4.2 Pressure Equalization - The microphone shall be placed in an altitude chamber and the pressure inside the chamber reduced to that corresponding to an altitude of 15,000 feet (approximately 16.7 inches of mercury). The response of the microphone shall also be measured at a frequency of 1000 Hz while the pressure is being varied at a rate of 5000 feet per minute (up or down). The sensitivity of the microphone at 1000 Hz shall not vary by more than ±3 db during the pressure cycling.

4.5.5 Temperature Extremes -

4.5.5.1 Low Temperature - Expose the microphone to an ambient temperature of -62°C for 2 hours. Remove the microphone from the cold chamber and immediately measure its response under normal room temperature conditions.

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4.5.5.2 High Temperature - Expose the microphone to an ambient temperature of +71°C for 2 hours. Remove the microphone from the hot chamber and immediately measure its response under normal room temperature conditions.

4.5.6 Salt Spray - The microphone shall be subjected to the salt spray test in accordance with Specification MIL-T-5422 for a period of 50 hours. The microphone shall be washed in tap water to remove excess salt deposits and then excess moisture removed.

*4.5.7 Temperature Cycling - The microphone shall be placed within the chamber and maintained for a period of at least 1 hour or until the microphone stabilizes at a temperature of 25° ±15°C. The chamber temperature shall then be reduced to -55°C and maintained at this condition for at least 1 hour or until the microphone stabilizes. The internal temperature of the chamber shall then be increased to 71°C and maintained at this condition for at least 1 hour or until the microphone stabilizes. The internal temperature shall then be returned to 25° ±15°C. This shall be repeated five times. The microphone shall be removed from the chamber and response measurements made after stabilization at room temperature.

4.5.8 Humidity - The microphone shall be subjected to the humidity test of Specification MIL-T-5422.

4.5.9 Distortion - The distortion of the microphone shall be measured for compliance with requirements of 3.5.3 by using a distortion analyzer connected across the 5.0 Ohm resistor of Figure 3.

4.5.10 Linearity - The linearity of the microphone shall be measured to determine compliance with requirements of 3.5.4.

4.5.11 Vibration - The microphone shall be subjected to the vibration test of Specification MIL-T-5422.

4.5.12 Stray Magnetic Field Measurements - Stray magnetic field measurements shall be made in a shielded room or in an area proved to the satisfaction of the cognizant Government Inspector to be free from appreciable magnetic disturbances. The microphone under test shall be rotated on its axis at various points on the surface of an 8 inch radius sphere surrounding a Keuffel and Esser Type No. 5600 compass or approved equivalent.

4.5.13 Impact Test - The microphone shall be subjected to 20 random drops, from a 6 foot height to a covered concrete floor. The covering shall consist of 1/8 or 3/16 inch thick asphalt tile thoroughly cemented to the concrete floor using a minimum thickness of cement.

4.5.14 Cord Test Methods - The cord shall be tested in accordance with Specification MIL-C-5898 to determine conformance with 3.4.1.3.3.

4.6 Life Tests - The microphone shall be subjected to a Life Test at the contractor's plant under the following simulated service conditions.

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Temperature	Normal room temperature
Altitude	Normal room
Humidity	Room Ambient

4.6.1 Test Data - The contractor shall keep a daily record as to the performance of the microphone, making particular note of any deficiencies or failures. In the event of part failures during this test, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

(1) Part Type Number.

(2) The circuit reference symbol-number.

(3) The part function.

(4) Name of the manufacturer.

(5) Nature of the failure.

(6) The number of hours which the part operated prior to failure.

4.6.1.1 Microphone Failure - In the event of a failure the Government Inspector shall be notified and the cause of the failure immediately determined. If the cause of the failure indicates material or design corrections are necessary, the Naval Air Systems Command shall determine if such corrective action is necessary prior to acceptance of additional microphones.

4.6.2 Test Duration - The duration of the Life Test shall be 300 hours.

4.6.2.1 Test Periods - The test may be run continously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically the microphone shall be turned on and off several times and put through all phases of its performance.

4.6.2.2 Performance Check - At approximately eight hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the Naval Air Systems Command.

4.6.3 Life Test Schedule - The microphone to be tested shall be representative of the current production in all significant aspects of design and manufacture. It shall be selected by the Government Inspector from microphones which have passed the Individual Tests. The Life Test shall be performed on the following quantities of microphones unless otherwise stated in the contract or approved by the Naval Air Systems Command.

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Quantity of Microphones Offered for Acceptance	Quantity to be Tested
First 25	1
Next 26 to 200	1
Next 201 to 500	1
Over 500	1

4.6.4 Reconditioning of Test Samples - Microphones which have been subjected to the Life Test shall be reconditioned as follows:

(1) On completion of the Life Test, the microphone shall be reworked by the contractor by replacing all "wear" items. The "wear" item shall be determined by agreement between the contractor and the Bureau of Naval Weapons.

(2) After reworking, the contractor may submit the microphone for acceptance under the contract.

*4.7 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests as required by both the Qualification Phase and the Production Acceptance (Sampling) Phase shall be conducted. Classification of failure shall be in accordance with MIL-STD-781 and AR-34.

*4.7.1 Qualification Phase - Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level E shall be used. The Accept-Reject Criteria for Test Plan II shall be used.

*4.7.2 Reliability Production Acceptance (Sampling) Phase Tests - The equipment, throughout production, shall be tested as outlined in MIL-STD-781 (as modified herein) under the section entitled "Production Acceptance (Sampling) Phase of Production Reliability Tests". Test level E shall be used.

*4.7.2.1 Lot Size for Sampling Phase - The equipments constructed during one month shall be one lot. The Accept-Reject criteria for Test Plan II shall be used to determine the length of the tests in each lot. Unless otherwise specified in the contract or order, no equipments shall be shipped until an Accept decision is reached under each lot.

*4.7.3 Test Details - The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, ect., shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests.

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4.8 Pre-Submission Testing - No item, part or complete microphone shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest -

4.9.1 Individual Tests - Each microphone which fails to meet all requirements set forth in 4.3.1 shall require immediate rejection.

*4.9.2 Sampling Tests - If any microphone selected fails to successfully pass all sampling tests, it shall cause rejection of the lot represented. Complete details concerning the cause of failure and steps taken to insure correction of such causes in future production shall be forwarded by the cognizant Government Inspector to the Naval Air Systems Command.

5. PREPARATION FOR DELIVERY

*5.1 General - All major items and parts of the microphone shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in the Specification MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in appendix D of MIL-STD-794.

6. NOTES

6.1 Use - It is intended that this microphone be used with standard Navy oxygen masks and when installed therein (with the oxygen mask properly attached to the human head), the over-all frequency response shall approach a flat characteristic between 300 Hz and 3000 Hz.

*6.2 Qualification - With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by this date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Naval Ammunition Depot, Crane, Indiana 47522, Code 3072 and information pertaining to qualification of products may be obtained from that activity.

6.3 Ordering Data - The procurement document should specify the following:

(1) Title, number and date of this Specification.

(2) Either a complete Microphone Dynamic, Oxygen Mask M-94B/A or any of the items of 1.2 as required.

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*6.4 Precedence - When the requirements of the contract, this specification, or applicable subsidiary specification are in conflict, the following precedence shall apply.

(1) Contract - The contract shall have a precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviations from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the Naval Air Systems Command.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

(4) Reordered Equipment - The following specifications shall take precedence over those concerning performance requirements listed or referenced in paragraph 2.1 where equipment is reordered.

MIL-E-5400P	2 July 1973
MIL-T-5422F	30 November 1971
MIL-I-6181D	25 November 1959
MIL-STD-781B	15 November 1967

*6.5 An asterisk preceding a paragraph number denotes a change made in that paragraph from the 'B' revision of this specification.

Project Number 5965-N133

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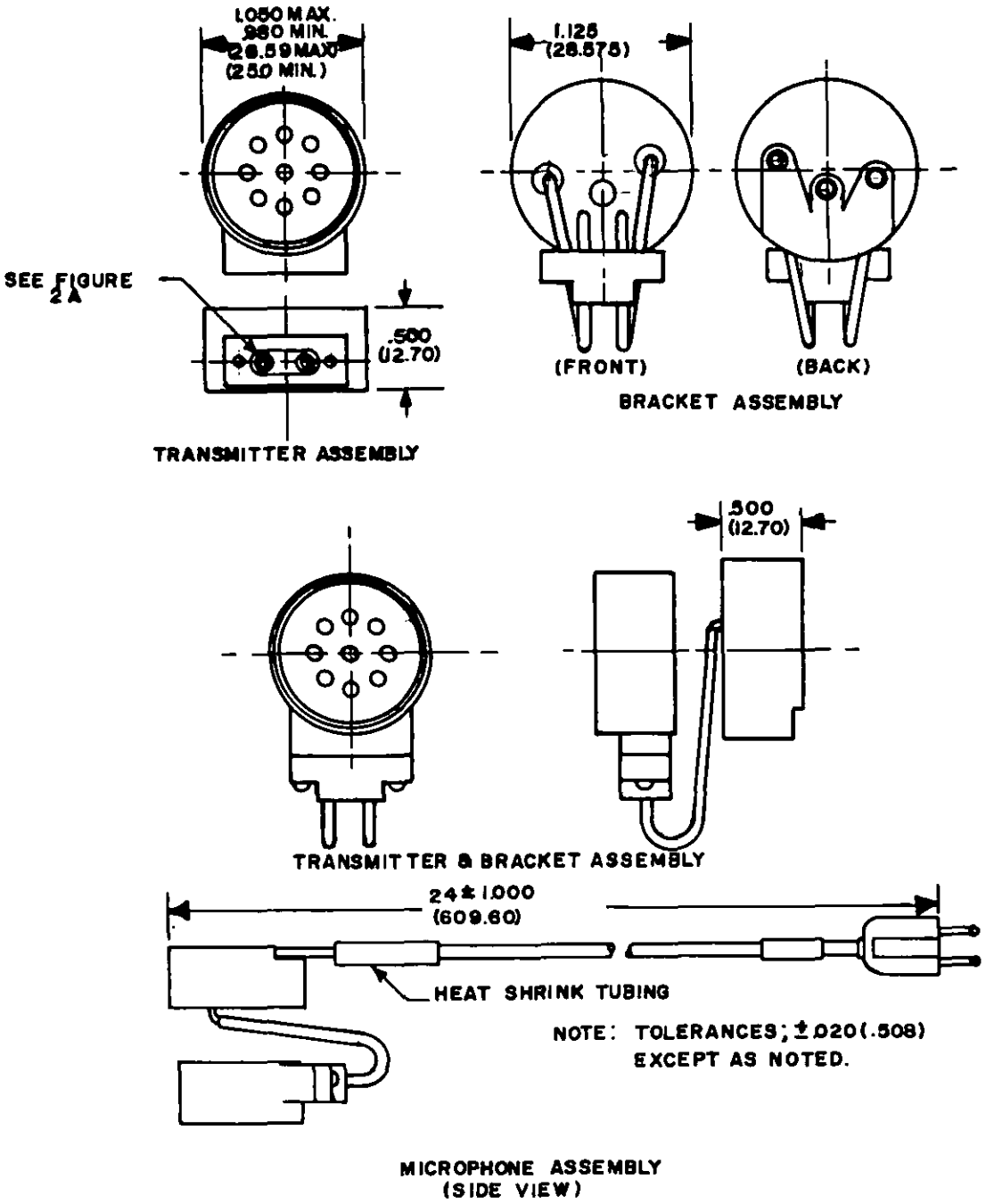
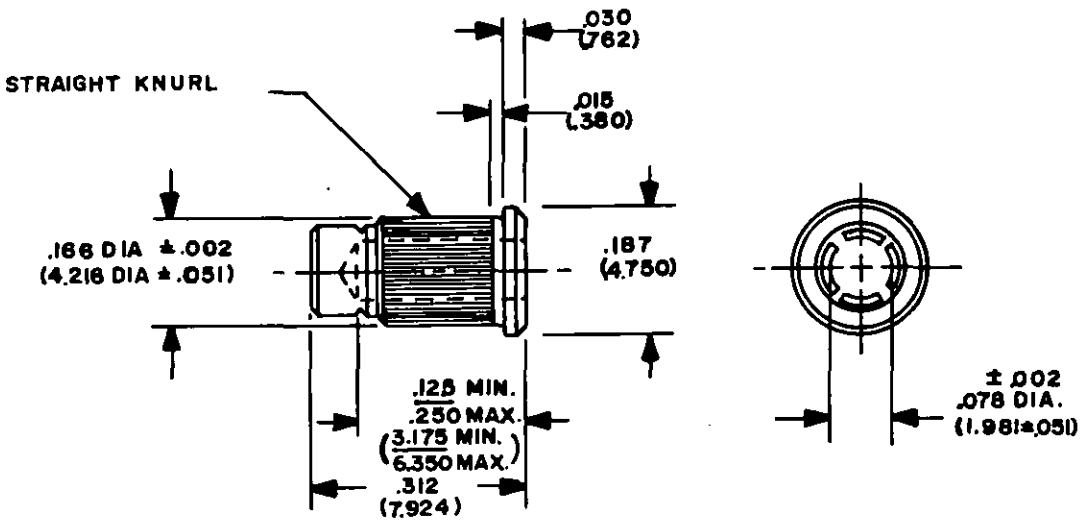


FIGURE 1

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NOTES:
1. MATL:
   Inner sleeve: Beryllium Copper 25, Per, QQ-C-530/A
   Outer sleeve: Half Hard Brass 22, Per, QQ-B-627
2. FINISH: (Outer Sleeve Only)
   Tin, .0002" Min. Per, MIL-T-10727, Type 1.
3. () Dimensions in Millimeters. All others in Inches.



CONNECTOR
FIGURE 2

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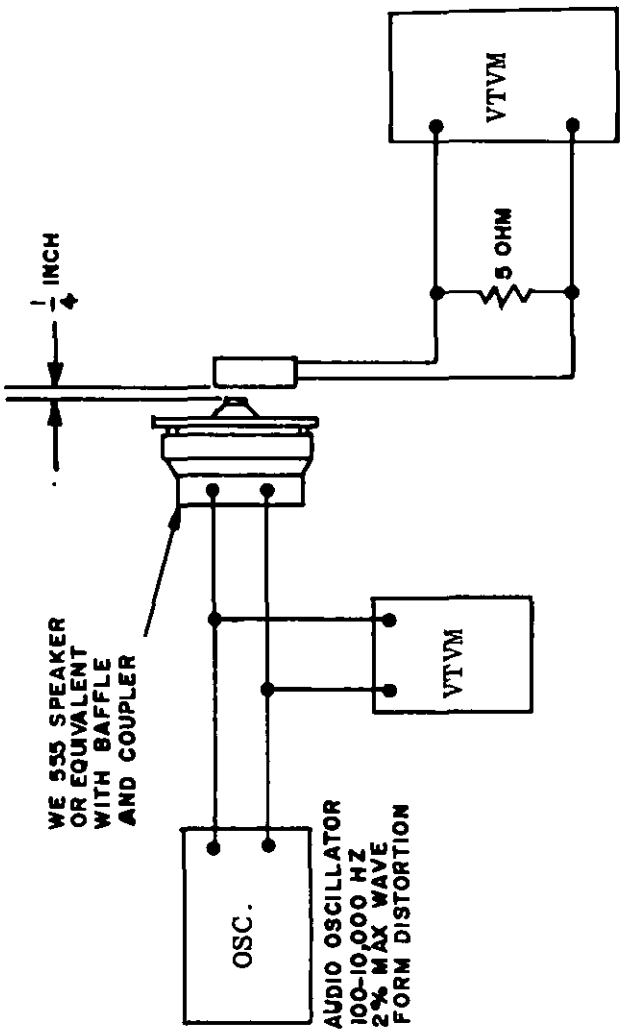
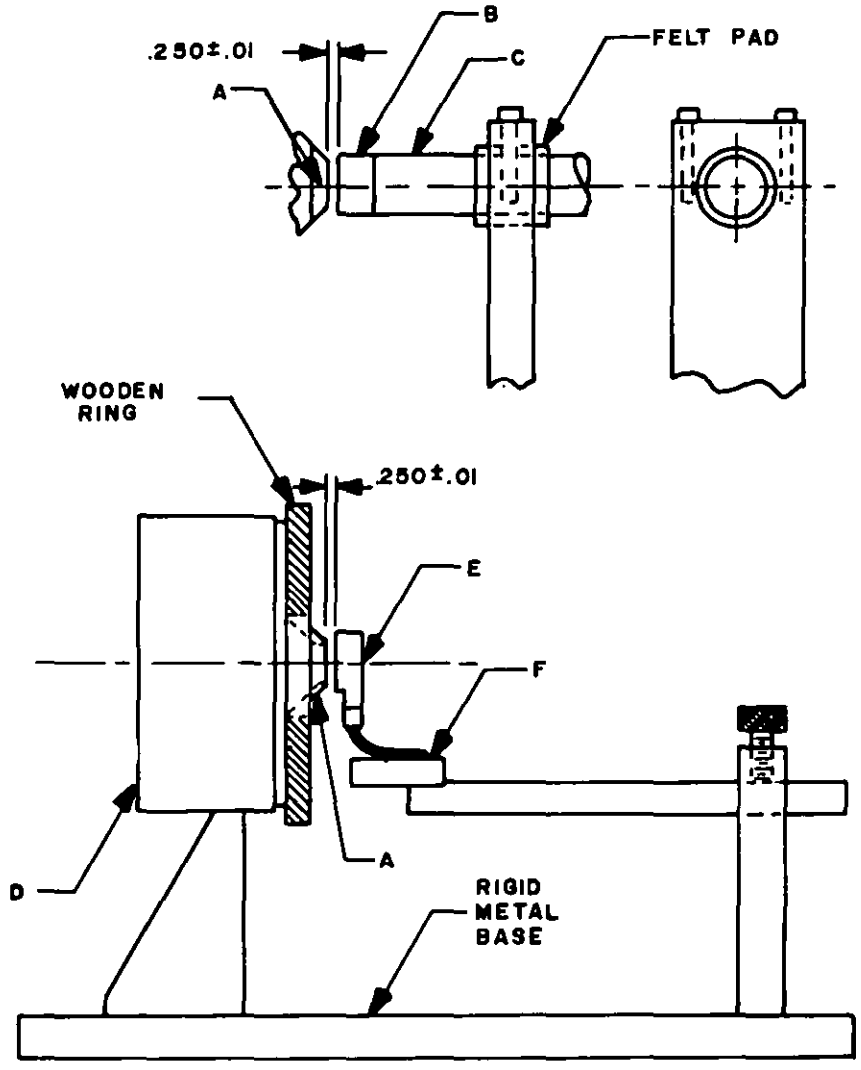


FIGURE 3 STANDARD TEST CIRCUIT

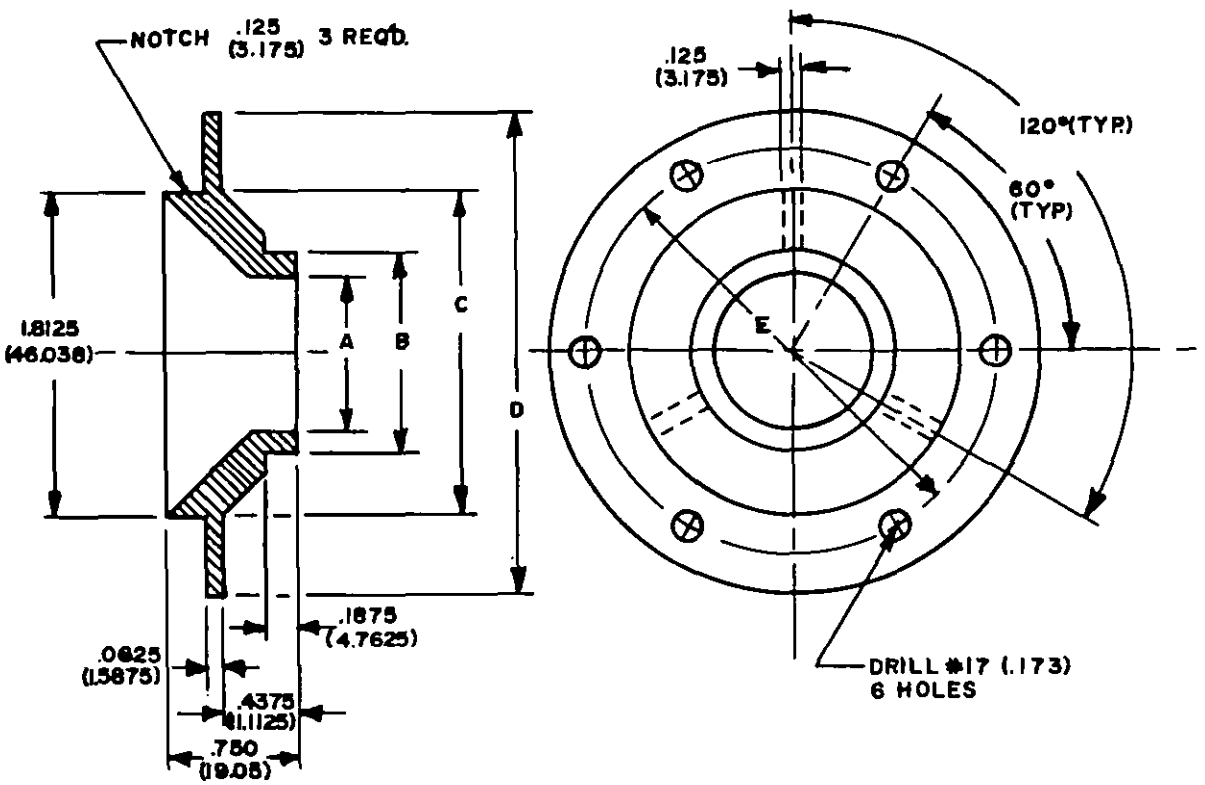
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A. MICROPHONE CALIBRATING COUPLER
B. W.E. 640-AA CONDENSER MICROPHONE, WITH GRID OR EQUIVALENT
C. MICROPHONE PREAMPLIFIER
D. SOUND SOURCE
E. MICROPHONE UNDER TEST
F. BRACKET ASSEMBLY

FIGURE 4

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DIMENSION	INCHES	(MILLIMETER)
A DIA.	.875	22.225
B DIA.	1.125	28.575
C DIA.	1.8125	46.0375
D DIA.	2.6875	68.2625
E DIA.	2.250	57.150

FIGURE 5

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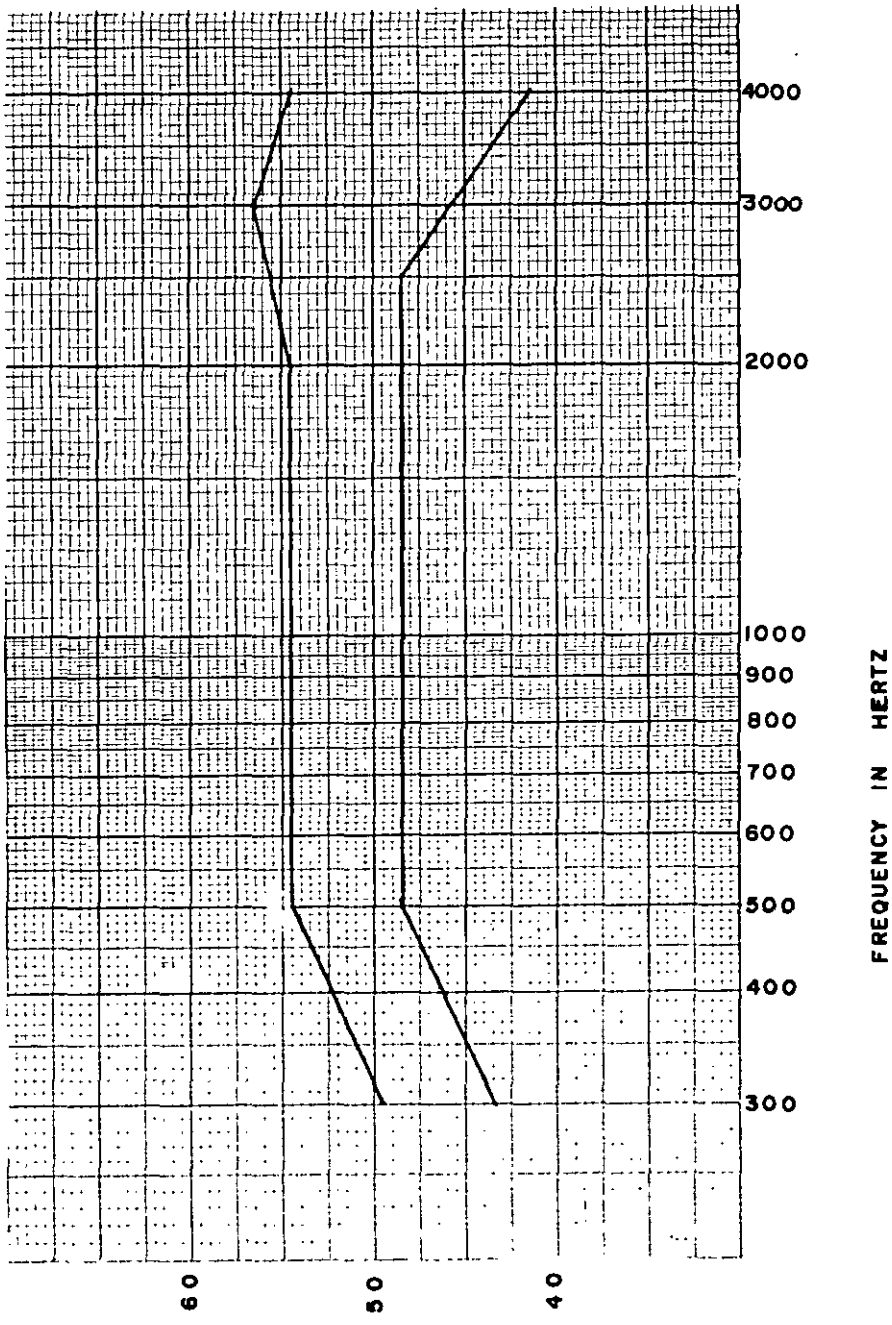


FIGURE 6

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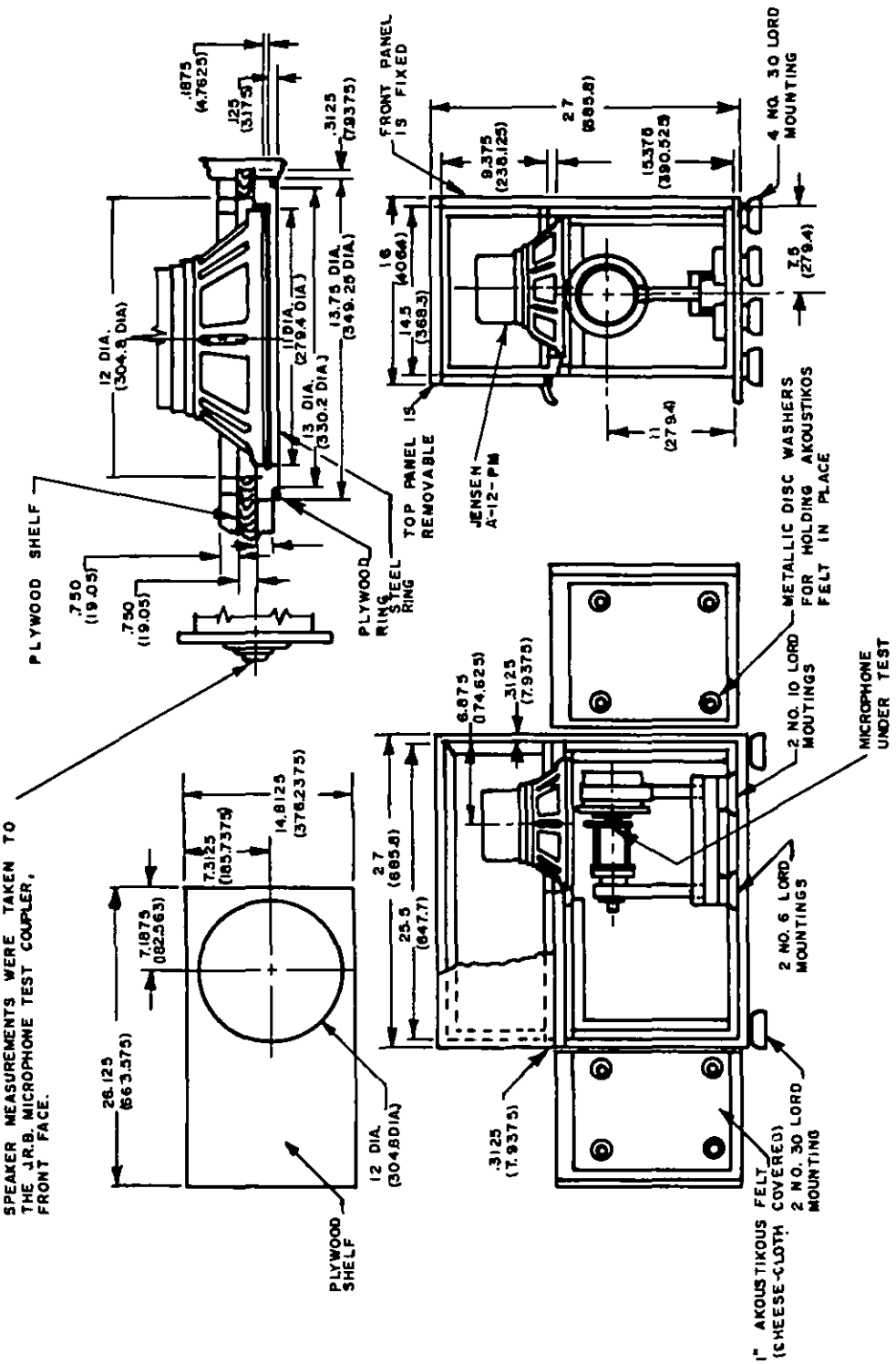


FIGURE 7

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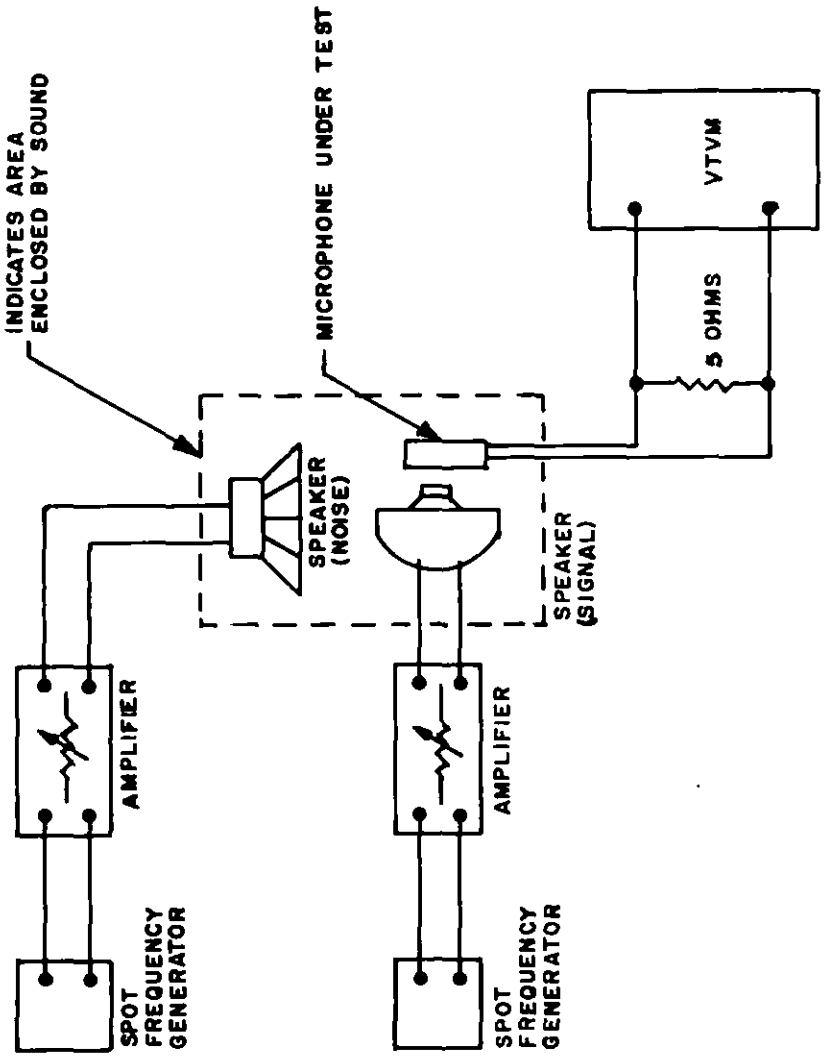
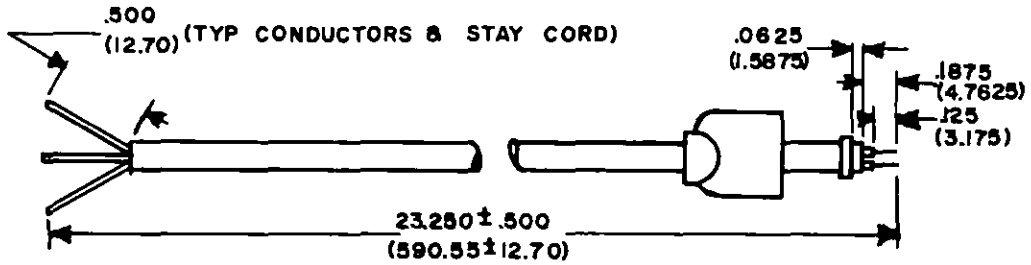
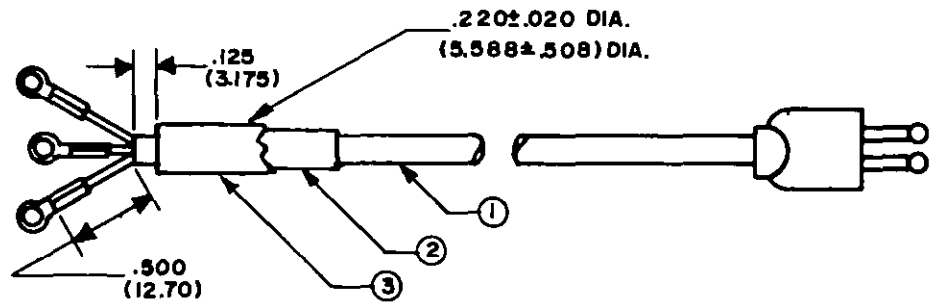


FIGURE 8

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NOTES:

1. HEAT SHRINK ITEM 2 ON ITEM 1, THEN HEAT SHRINK ITEM 3 ON ITEM 2. CAUTION; DO NOT HEAT SHRINK BOTH ITEM 2 & 3 AT SAME TIME.
2. ATTACH STAYCORD TERMINAL AS CLOSE AS POSSIBLE TO END OF OUTER JACKET.

FIGURE 9

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-M-21734C(AS) Microphone, Dynamic, Oxygen Mask M-94B/A

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED S/N 0102-014-1802

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